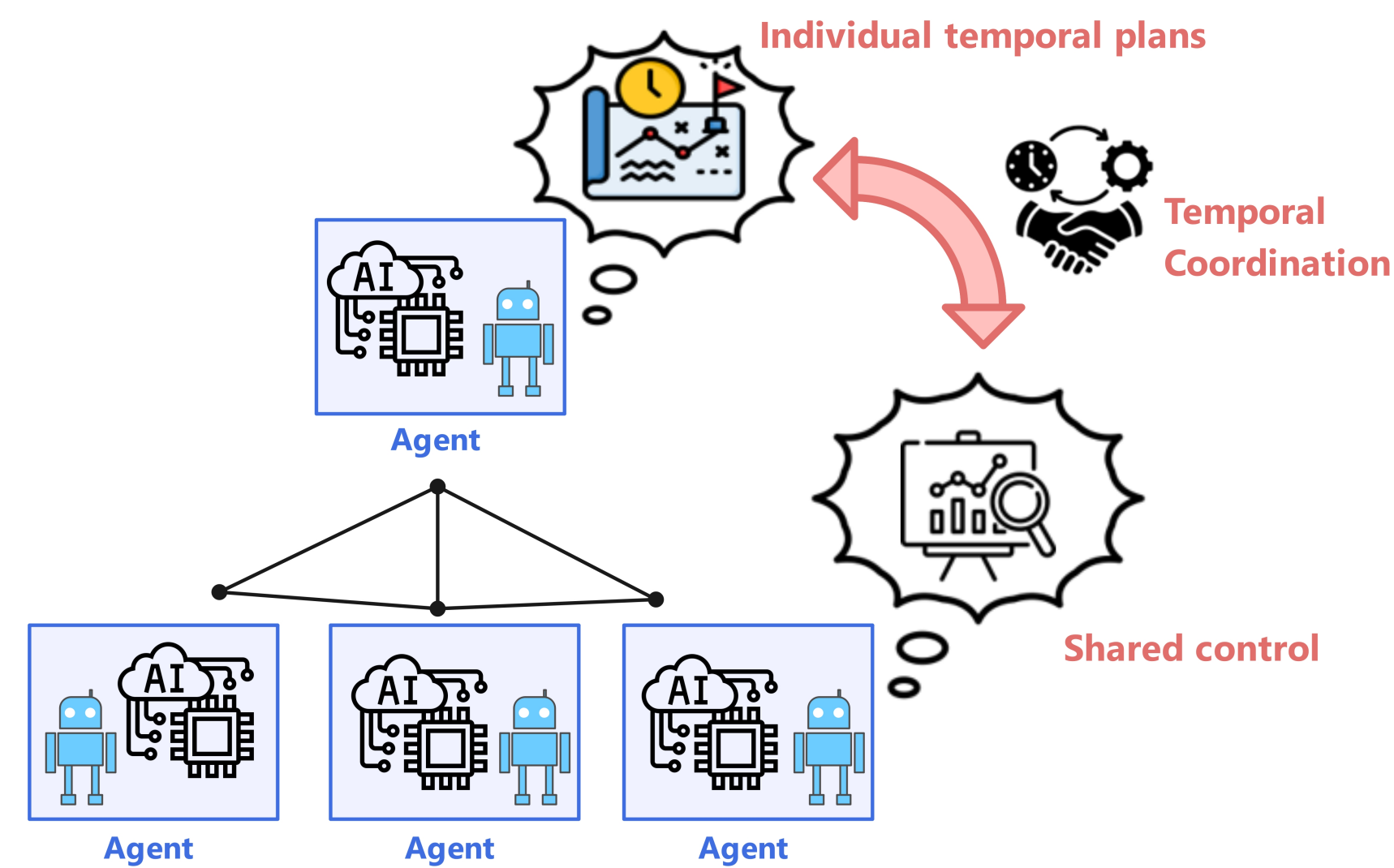


Centralized and Distributed Approaches for Restoring Weak Controllability of Multi-Agent Interdependent STNUs

Introduction

In multi-agent temporal planning, agents first build individual plans through task allocation and planning (possibly via negotiation) and then must check and coordinate their temporal executability. We focus on this second stage in a setting where the task owner controls task durations, while other compliant agents may depend on them and learn their values only near or during execution. Besides checking executability, such agents may **repair** a non-executable plan by negotiating shared constraints.



Background

A commonly used model for checking temporal consistency of plans is the Simple Temporal Network (STN) by finding an assignment of real-value variables called timepoints that satisfies a set of constraints expressing sets of possible durations with convex intervals of values relating them. STNUs extend STNs with uncertain (*contingent*) durations and replace consistency with *controllability*: a plan is controllable if an execution strategy exists regardless of contingent outcomes. **Weak Controllability (WC)** assumes contingent duration will be known just before plan execution. Hence, an STNU is WC if a solution exists for each possible combination of contingents' durations.



Multi-agent Interdependent Simple Temporal Network with Uncertainty

Multi-agent Interdependent STNUs (MISTNU) model extends STNUs to multi-agent systems (one STNU per agent). MISTNUs introduced a semantically distinct notion of constraints called *contract*, corresponding to a shared constraint being controllable for one agent but contingent for others. An MISTNU is Weak Controllable if each network (STNU) is WC.

Weak Repair (WR) : An MISTNU is Weak Repairable if tightening the bounds of contracts yield a Weak Controllable MISTNU.

Healthcare example

Consider a radiologist, a nurse, and a doctor. The radiologist performs an X-ray on the first patient, represented by contract p , whose duration is observed by the nurse and the doctor to be contingent: the nurse waits for the X-ray before escorting the patient, while the doctor can analyze the results only after its completion. The nurse then escorts the current patient and brings a second patient to the radiologist, represented by contract q , which is owned by the nurse but observed by the radiologist. Finally, contract r represents the second X-ray, along with the following dependencies involving the nurse and the radiologist. After this second X-ray, the radiologist must clean and inspect the equipment within the required time window.

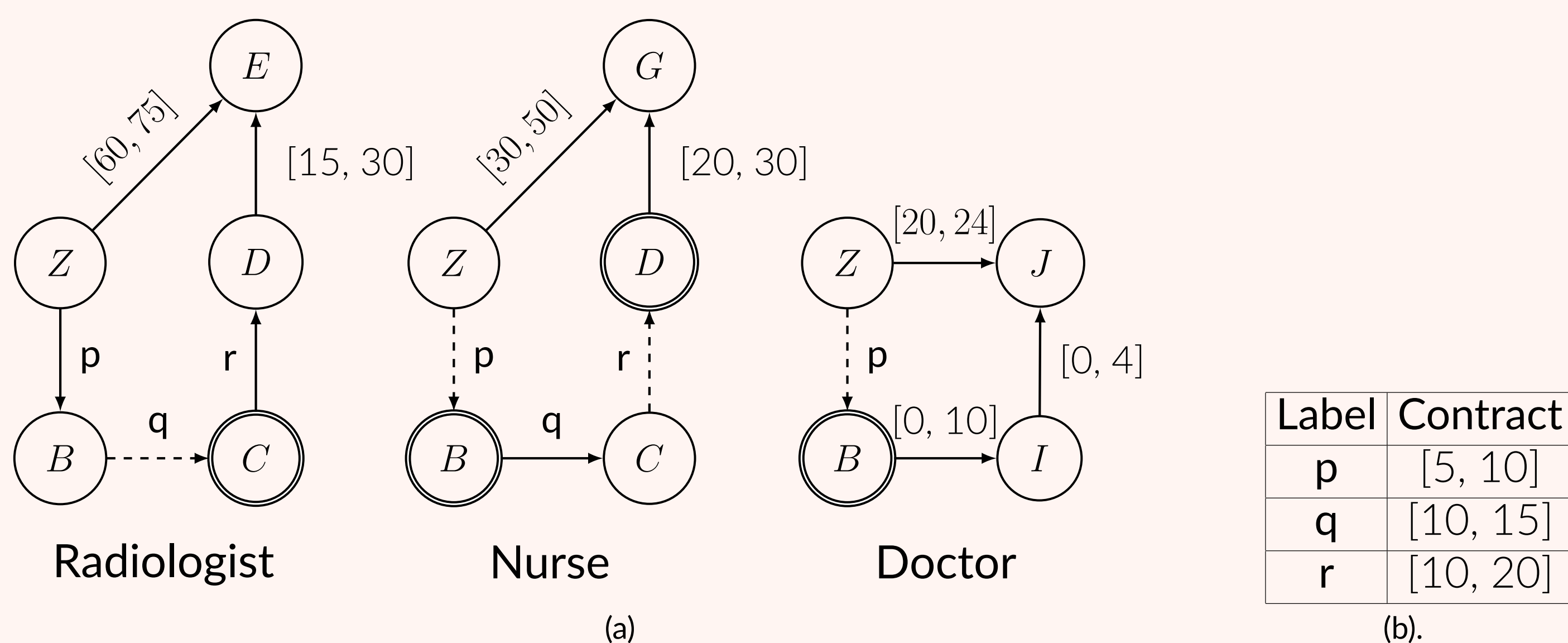


Figure 2. The corresponding MISTNU of the given healthcare example

Research questions

How can we efficiently repair a non WC MISTNU ?

Label	Contract
p	[5, 10]
q	[10, 15]
r	[10, 20]

Table 1. MISTNU not WC

Label	Contract
p	[10, 10]
q	[10, 10]
r	[10, 10]

Table 2. Repaired MISTNU

A Centralized Linear Programming Approach

Checking Weak Controllability is done by finding inconsistent cycles. Repairing non-WC MISTNUs is equivalent as expressing all inconsistent cycles as a set of linear constraints to satisfy.

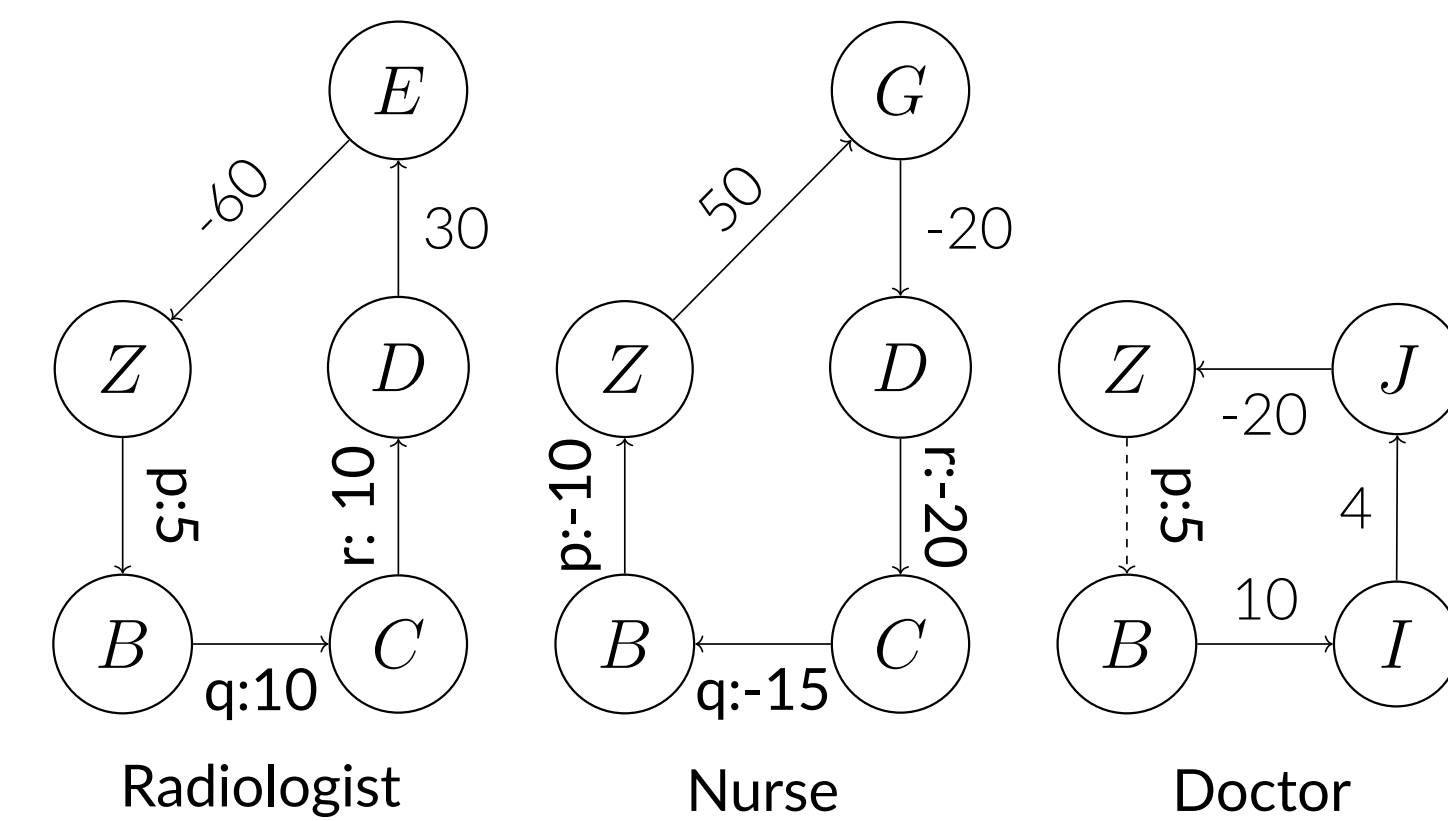


Figure 3. Inconsistent cycles of the MISTNU. STNU (a) is not WC as there is a cycle of length -5.

Encoding an inconsistent cycle

$$\text{(Radiologist)} \quad l^p - 5 + l^q - 10 + l^r - 10 \geq 5$$

$$\text{(Nurse)} \quad 10 - u^p + 15 - u^q + 20 - u^r \geq 15$$

$$\text{(Doctor)} \quad l^p - 5 \geq 1$$

A Distributed Approach

We model the Weak repair problem as a Distributed Constraint Optimization Problem (DCOP). The variables are the bounds of contracts, while constraints to satisfy are the set of inconsistent cycle transformed as weighted constraints.

$$\text{(Radiologist)} \quad 0 \text{ if } l^p - 5 + l^q - 10 + l^r - 10 \geq 5 \text{ else } +\infty$$

$$\text{(Nurse)} \quad 0 \text{ if } 10 - u^p + 15 - u^q + 20 - u^r \geq 15 \text{ else } +\infty$$

$$\text{(Doctor)} \quad 0 \text{ if } l^p - 5 \geq 1 \text{ else } +\infty$$

Minimizing the tightening of the bounds of contracts

$$\text{(Centralized)} \quad \text{Min} \left((u^p - l^p) + (u^q - l^q) + (u^r - l^r) \right)$$

$$\text{(DCOP)} \quad \left\{ u^p - l^p ; u^q - l^q ; u^r - l^r \right\} \rightarrow \text{Additional constraints}$$

Experimentation

The benchmark is organized into seven groups of 50 instances, with network sizes in (20, 30, 50, 80, 100) timepoints per agent network. For each size, the number of owned contracts per agent is fixed at 1-7.

Network size	20	30	50	80	100
Average # cycles	6±2	52±17	75±20	1045±300	7542±1800
Average arity	2±1	3±2	4±2	5±2	7±2

Table 3. Average number and arity of inconsistent cycles.

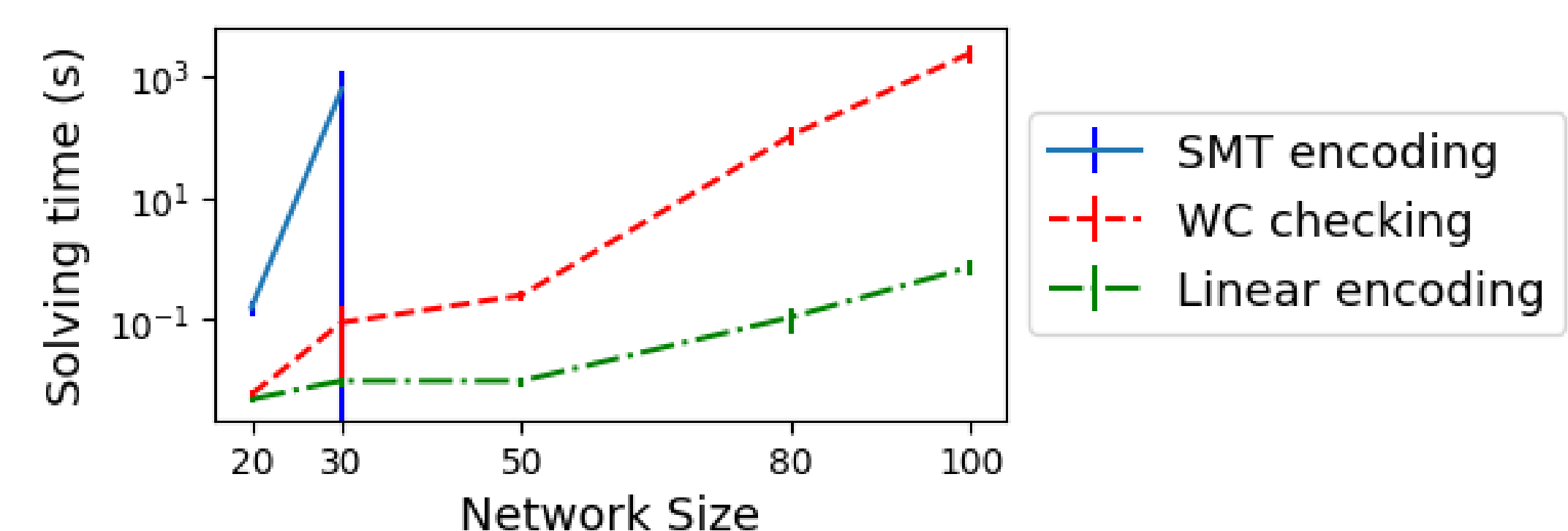


Figure 4. Centralized results. Average execution time, with standard deviation, as a function of network size. The plot separates WC checking, linear repair, and SMT repair. SMT repair exhibits high variance.

Metric	DPOP	ADOPT	AFB	SyncBB
Instances solved	55/250	138/250	250/250	250/250
Average time (s)	3.5±13.2	12.7±28.8	1.12±2.2	0.74±1.4

Table 4. SAT-version results for complete DCOP algorithms.

Network size	# solved instances		Average execution time (s)		Solution quality (AFB/SyncBB)
	AFB	SyncBB	AFB	SyncBB	
20	50/50	50/50	55.4±131.2	87±167.5	-
30	17/50	10/50	941.4±825.9	1155.6±942.6	75%
50	12/50	7/50	405.3±372.9	482.3±422.7	72%
80	0/50	0/50	-	-	82%
100	0/50	0/50	-	-	90%

Table 5. Optimal-version results for AFB and SyncBB algorithms with their solution quality on the SAT-version.